SmartInvestor — AI Voice Agent for SmartCrowd Dashboard

Context

The user has an Associate Product Manager interview at **SmartCrowd** tomorrow and wants to apply for the **ElevenLabs Customer Success** role. They want to build a project that demonstrates:

- **Builder mentality** — shipping a working product, not just talking about ideas
- **SmartCrowd domain knowledge** — understanding fractional real estate investing in MENA
- **ElevenLabs platform competence** — hands-on experience building and deploying a voice agent
- **Product thinking** — solving a real user problem (investor portfolio Q&A + education)

User profile: Non-technical PM based in Middle East, 2-3 days to build. Claude Code does the coding.

What We're Building

A **SmartCrowd dashboard clone** with an embedded ElevenLabs voice agent. The UI replicates the look and feel of SmartCrowd's actual property browse page (sidebar nav, property cards with tabs for Live/Funded/Exited). The voice agent sits as a floating assistant button and can answer two types of questions:

1. **Property-specific queries** — "What's the renovation progress on the Palm Jumeirah Flip?", "What's the rental yield for the 1-bedroom in IMPZ?", "Which property has the best ROI?"
2. **General knowledge queries** — "How does fractional investing work?", "What are SmartCrowd's fees?", "What's the difference between Hold and Flip?"

All property data is real data captured from SmartCrowd's platform API (1 live + 165 funded + 67 exited = 233 properties).

Architecture

```
┌──────────────────────────────────────────────┐
│   Next.js Web App (Vercel)                   |
|                                              |
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|  | Sidebar  |  | Property Browse View     |  |
|  | Nav      |  | ┌────────────────────┐   |  |
|  | - Explore|  | | Tabs: Live|Funded|   |  |
|  | - Wallet |  | |       Exited       |   |  |
|  | - Port-  |  | ├────────────────────┤   |  |
|  |   folio  |  | | Property Cards Grid |  |  |
|  | - Cart   |  | | (real SC data)     |   |  |
|  |          |  | └────────────────────┘   |  |
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|                                              |
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|  | 🎙 Floating Voice Agent Button       |    |
|  | → Opens voice conversation overlay   |    |
|  | → ElevenLabs React SDK (WebRTC)      |    |
```

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│                                                 │
│   ┌──────────────────────────────────────────┐ │
│   │ API Routes                                │ │
│   │ - /api/properties (lookup by ID/area/type)│ │
│   │ - /api/calculate-roi                      │ │
│   │ - /api/agent-token (ElevenLabs auth)      │ │
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│   ElevenLabs Agent (Cloud)                      │
│                                                 │
│   Voice: Professional, warm                     │
│   Knowledge: SmartCrowd general info            │
│   Tools:                                        │
│   - lookup_property (by ID, area, or type)      │
│   - search_properties (filter by criteria)      │
│   - calculate_roi (investment projections)      │
│   - get_renovation_status (for Flip props)      │
│   Persona: SmartCrowd Investment Advisor        │
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```

Property Data (Scraped from SmartCrowd)

Funded Properties (20 — sample shown)

ID	Name	Type	Area	Beds	Sqft	Yield/ROI	Purchase	Market Value
SC-330	Studio, Discovery Gardens	Hold	Discovery Gardens	-	550	6.07%	578K	580K
SC-326	1B+Study, Bali Residences JVT	Hold	JVT	1	824	5.94%	1,290K	1,300K
SC-325	3BR Penthouse, Bellevue Downtown	Flip	Downtown	3	3016	15.01% ann.	-	-
SC-323	1B Lakeside, IMPZ	Hold	IMPZ	1	596	6.25%	600K	600K
SC-319	Old Town Island, Downtown	Flip	Old Town	2	2227	15.33% ann.	-	-

SC-317	Serenia West, Palm Jumeirah	Flip	Palm Jumeirah	3	3447	15.49% ann.	-	-
SC-315	1BR Hera Tower, Sports City	Hold	Sports City	1	760	6.72%	835K	850K
SC-314	1BR Joya Blanca, Arjan	Hold	Arjan	1	746	6.14%	995K	1,100K
SC-309	5BR Villa, Sidra 3 Dubai Hills	Flip	Dubai Hills	5	4109	15.19% ann.	-	-
SC-312	3BR Penthouse, Downtown	Flip	Downtown	3	2800	15.47% ann.	-	-

Exited Properties (20 — sample shown)

ID	Name	Type	Area	Exit Price	Total Rental	Holding	ROI
SC-285	3BR Villa, Palm Jumeirah	Flip	Palm Jumeirah	24,000,000	-	14 mo	+24.59%
SC-282	3BR Duplex, DIFC	Flip	DIFC	8,700,000	-	18 mo	+25.01%
SC-231	1BR Duplex, Fortunato JVC	Hold	JVC	894,469	130,228	34 mo	+49.28%
SC-208	1BR, 8 Blvd Walk Downtown	Hold	Downtown	1,531,158	178,605	43 mo	+51.12%
SC-179	1BR Fortunato Tower, JVC	Hold	JVC	713,403	102,376	35 mo	+63.92%
SC-187	Yacht Bay Studio, Marina	Hold	Dubai Marina	910,307	165,191	47 mo	+47.08%

Data Capture Summary

All property data has been captured via API interception from SmartCrowd's public API (`api.phoenix.smartcrowd.ae`):

- **1 Live**: SC-327 "Villa in the Sky" Penthouse, DIFC (Flip, limited public data)

- **165 Funded**: Full structured data including rental yields, market values, renovation progress, investor counts, funded dates
- **67 Exited**: Full data including exit prices, total rental income, holding periods, ROI percentages

Raw data file: `/Users/flashy/.claude/projects/-Users-flashy/0eaeaf78-3e8c-4ae2-88c0-44cb5cf92e30/tool-results/mcp-playwright-browser_run_code-1771926478771.txt`

Each property includes: id, code, title, investmentType, price, propertyType, investmentCategory, propertyStatus, renovationProgress, rental (yields, rent, income), performance (investors, ROI, funded%), valuation (market value, sale proceeds), physical (beds, baths, sqft), location (area, building, city), auction (dates)

(Full dataset of all 233 properties will be in `src/data/properties.ts`)

Agent Knowledge Base

General SmartCrowd Knowledge (baked into system prompt)

- **Platform**: DFSA-regulated, DIFC-registered, AED 500 minimum, 5000+ investors, 165+ funded properties, 67 exits
- **How it works**: Register (5 min) → KYC (free) → Fund wallet → Browse → Invest → Earn returns → Exit
- **Hold model**: 3-5yr hold, earn rental income + capital appreciation, target 7-11% annual
- **Flip model**: 9-12 month, buy undervalued → renovate → sell, target 15-30% ROI
- **Fees**: 1.5% entry, 0.5% annual admin, 2.5% exit, 0% performance fee
- **Track record**: 67 exits, avg ~35-40% net ROI, AED 220M+ in exit proceeds
- **Exit options**: 5-year investor vote, Share Transfer Facility (March/September windows)
- **Advantages**: Lowest entry (AED 500), full lifecycle management, no performance fees, Sharia-compliant

Property-Specific Knowledge (via tool calls)

The agent uses tools to query the property database, enabling questions like:

- "What's the renovation progress on the Sidra villa?" → calls `get_renovation_status("SC-309")` → "80% complete"
- "What's the yield on the IMPZ properties?" → calls `search_properties({area: "IMPZ"})` → lists SC-323, SC-321, SC-320
- "If I invest 10,000 in the Sports City property, what would I earn?" → calls `calculate_roi({amount: 10000, property_id: "SC-315"})`

Implementation Plan

Phase 1: Setup (Day 1, ~1 hour)

1. **Create ElevenLabs account** at elevenlabs.io → generate API key
2. **Scaffold Next.js project** at `~/Projects/smart-investor/`
   - `npx create-next-app@latest` with App Router + Tailwind
   - Install: `@11labs/react`, `elevenlabs`, shadcn/ui
   - Init shadcn: `npx shadcn@latest init`

Phase 2: Build Data Layer (Day 1, ~1 hour)

1. **`src/data/properties.ts`** — All 233 properties (1 live + 165 funded + 67 exited) as typed objects
   - Transform raw API data from captured file into clean TypeScript interfaces
   - Source: `/Users/flashy/.claude/projects/-Users-flashy/0eaeaf78-3e8c-4ae2-88c0-44cb5cf92e30/tool-results/mcp-playwright-browser_run_code-1771926478771.txt`
   - Fields per property: code, title, investmentType, price, area, beds, sqft, rentalYield, marketValue, renovationProgress, investors, rentalIncome, ROI data, dates
2. **`src/data/knowledge.ts`** — SmartCrowd constants (fees, stats, FAQs as structured data)

Phase 3: Build Dashboard UI (Day 1-2, ~3-4 hours)

1. **Layout** (`app/layout.tsx`)
   - SmartCrowd-style sidebar: logo, nav items (Explore, Wallet, Portfolio, Cart, Notifications), light/dark mode toggle
   - Main content area to the right
2. **Property Browse Page** (`app/page.tsx`)
   - Tab bar: Live (1) | Funded (165) | Exited (67) — counts from actual data
   - Sort & Filter controls (area dropdown, type filter)
   - Scrollable property cards grid with all 233 properties, filtered by active tab
3. **Property Card Generation** (`src/components/PropertyCard.tsx`)
   - Single component that renders differently based on property type/status
   - Receives a typed property object from `properties.ts` and adapts layout:
   - **Hold cards** (funded): property image (from `baseImageUrl` in API data), Hold badge, tag badges (e.g. "Instant Returns", "High Yield"), investor count, property name, SC-ID code, beds/sqft/area/rental type, rental yield %, purchase price, current market value with change %, funded date, rental income to date
   - **Flip cards** (funded): property image, Flip badge, investor count, property name, SC-ID code, beds/sqft/area, annualized ROI %, estimated timeline, renovation progress bar (0-100%), project cost
   - **Exited cards**: property image, Hold/Flip badge, investor count, property name, SC-ID code, beds/sqft/area/rental type, exit price, total rental income, holding period, total return ROI amount + percentage
   - **Live card** (SC-327): Flip badge, title only with "Sign up to unlock" overlay (mimics real SC behavior)
   - Images: use `baseImageUrl` from captured API data where available, fallback to a gradient placeholder
4. **`src/components/PropertyGrid.tsx`** — maps over the filtered properties array and renders a `PropertyCard` for each, with infinite scroll or virtualized list for performance (165 cards on funded tab)
5. **shadcn components needed**: `button` , `card` , `badge` , `tabs` , `dialog` , `progress` , `separator` , `scroll-area`

Phase 4: Configure ElevenLabs Agent (Day 2, ~1-2 hours)

1. **Create agent in ElevenLabs dashboard**
   - Name: "SmartInvestor"
   - Voice: Professional, warm (test a few options)
   - System prompt: Full SmartCrowd knowledge base + instructions for personality and tool usage
   - First message: "Hi! I'm your SmartCrowd investment advisor. I can help you understand our properties, explain how fractional investing works, or answer any questions about your

portfolio. What would you like to know?"

2. **Register 4 tools** (client-side via the React SDK):
   - `lookup_property` : params `{property_id: string}` → returns full property details
   - `search_properties` : params `{area?: string, type?: "Hold"|"Flip"|"Exited", min_yield?: number}` → returns matching properties
   - `calculate_roi` : params `{investment_amount: number, property_id: string, holding_years?: number}` → returns projected returns after fees
   - `get_renovation_status` : params `{property_id: string}` → returns renovation % and timeline for Flip properties
3. **Test in playground** — iterate on prompt until natural

Phase 5: Build Voice Agent Component (Day 2, ~2 hours)

1. **`src/components/VoiceAgent.tsx`** — Main conversation component
   - Uses `@11labs/react` `useConversation` hook
   - Floating mic button (bottom-right corner of dashboard)
   - When clicked: opens overlay/modal with:
     - Pulsing animation while agent speaks
     - "Listening..." / "Speaking..." / "Thinking..." states
     - End call button
   - Client-side tool handlers that query the local property data (no API routes needed — data is in-memory)
2. **`app/api/agent-token/route.ts`** — Server-side route to generate signed URL for ElevenLabs (keeps API key server-side)

Phase 6: Integration & Polish (Day 2-3, ~2 hours)

1. **Wire up tool handlers** — when agent calls `lookup_property`, the client-side handler searches the local properties array and returns results
2. **Polish the dashboard**
   - Match SmartCrowd's blue/white color scheme
   - Property card hover states
   - Responsive for mobile demo
   - Smooth transitions on tab switch
3. **Test conversations end-to-end**:
   - "What's the rental yield for the 1-bedroom in IMPZ?" → agent calls search_properties → "The 1-bedroom in Lakeside IMPZ (SC-323) has a rental yield of 6.25%..."
   - "How's the renovation going on the Sidra villa?" → agent calls get_renovation_status → "The 5-bedroom villa in Sidra 3 is at 80% renovation progress..."
   - "How does fractional investing work?" → agent answers from knowledge base
   - "What fees does SmartCrowd charge?" → agent explains 1.5% entry, 0.5% annual, 2.5% exit
   - "If I invest 5000 dirhams in the Sports City property, what would I make?" → agent calls calculate_roi → returns projection

Phase 7: Deploy (Day 3, ~1 hour)

1. **Init git repo + push to GitHub**
2. **Deploy to Vercel** — set env vars (ELEVENLABS_API_KEY, NEXT_PUBLIC_AGENT_ID)
3. **Test live URL** on phone + desktop
4. **Record a 60-second demo video** (optional but high-impact for ElevenLabs application)

File Structure

```
smart-investor/
├── app/
│   ├── layout.tsx                # Dashboard layout with sidebar
│   ├── page.tsx                  # Property browse page (tabs + grid)
│   ├── globals.css               # Global styles + Tailwind + SC color scheme
│   └── api/
│       └── agent-token/route.ts  # ElevenLabs signed URL generation
├── src/
│   ├── components/
│   │   ├── Sidebar.tsx           # SmartCrowd-style sidebar nav
│   │   ├── PropertyTabs.tsx      # Live/Funded/Exited tab bar
│   │   ├── PropertyCard.tsx      # Card component (adapts for Hold/Flip/Exited)
│   │   ├── PropertyGrid.tsx      # Grid of property cards
│   │   ├── VoiceAgent.tsx        # ElevenLabs voice agent (floating button + overlay)
│   │   └── ui/                   # shadcn components
│   ├── data/
│   │   ├── properties.ts        # All 233 properties (1 live + 165 funded + 67
exited)
│   │   └── knowledge.ts         # SmartCrowd constants, fees, FAQs
│   ├── lib/
│   │   ├── utils.ts             # shadcn cn() utility
│   │   └── agent-tools.ts      # Tool handler functions for ElevenLabs agent
│   └── types/
│       └── property.ts          # TypeScript types for properties
├── .env.local                   # ELEVENLABS_API_KEY, NEXT_PUBLIC_AGENT_ID
├── next.config.ts
├── tailwind.config.ts
├── components.json              # shadcn config
├── package.json
└── tsconfig.json
```

Key Dependencies

- `next` (15+) — React framework with App Router
- `@11labs/react` — ElevenLabs React SDK for voice conversations
- `elevenlabs` — ElevenLabs JS SDK (server-side token generation)
- `tailwindcss` — Styling
- shadcn/ui: `button`, `card`, `badge`, `tabs`, `dialog`, `progress`, `separator`

Verification

1. **Dashboard renders**: `npm run dev` → sidebar + property cards display correctly across all 3 tabs
2. **Data accuracy**: Property cards match real SmartCrowd data (spot-check SC-315 yield = 6.72%, SC-309 renovation = 80%)
3. **Voice agent connects**: Click floating mic → microphone permission → agent greets you
4. **Property queries work**: Ask "What's the yield on the IMPZ property?" → agent returns 6.25%
5. **Renovation queries work**: Ask "How far along is the Sidra villa renovation?" → agent returns 80%

6. **General knowledge works**: Ask "How does SmartCrowd work?" → coherent explanation
7. **ROI calculator works**: Ask "If I invest 10,000 in Sports City, what do I earn?" → returns projection with fee breakdown
8. **Deploy test**: Live URL works on mobile and desktop
9. **Demo readiness**: Can walk through dashboard + have a voice conversation in under 2 minutes

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What This Demonstrates

SmartCrowd Interview (tomorrow)

- "I built a voice-powered assistant for your dashboard that can answer investor questions about their properties and about fractional investing in general"
- Shows: deep product understanding, initiative, builder mentality, AI product vision
- Talking point: "This could reduce Help & Support tickets by 40%+ and improve investor confidence during onboarding"

ElevenLabs Application

- Built a production-quality conversational agent on the ElevenLabs platform with:
  - Custom tool integrations (4 tools)
  - React SDK + WebRTC integration
  - Real enterprise use case (FinTech investor dashboard)
  - Knowledge base + dynamic data retrieval
- Shows: hands-on platform expertise, enterprise customer empathy, ability to advise on agent design
- Talking point: "I built this to understand the platform from a customer's perspective — the biggest insight was how tool design determines conversation quality"